

The Financial Impacts of Climate Shocks on Healthcare

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Dissertation Overview and Contributions

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Synopsis. This dissertation investigates how climate shocks propagate through U.S. health-care financing on both the demand (households/insurers) and supply (hospitals) sides, and then embeds those empirical relationships in a structural policy model. Across three chapters, it (i) estimates the causal effects of extreme heat, wildfire smoke/PM_{2.5}, and disaster events on insurance pricing and household financial distress; (ii) quantifies consequences for hospital operating performance and balance-sheet risk; and (iii) integrates these margins to evaluate counterfactual policy instruments - including climate-risk adjusters, geographically targeted premium subsidies, disaster-triggered cost-sharing waivers, and provider stabilization mechanisms. The unifying contribution is to connect environmental risk to core health-economics objects - prices, financial protection, and provider solvency-within a coherent empirical-structural pipeline.

Chapter 1: Climate Shocks, Insurance Premiums, and Medical Debt (Demand Side)

High-level summary. Chapter 1 links local climate hazards to two financial outcomes: benchmark Affordable Care Act marketplace premiums at the rating-area level and medical debt in collections at the county level. The chapter assembles a national panel that aligns exposure timing with premium-setting and billing calendars, and estimates stacked event-study and triple-difference designs to net out policy, macroeconomic, and seasonal confounders.

Contribution to the literature. The chapter provides the first joint elasticities of premiums and household debt with respect to climate shocks, bridging environmental economics, household finance, and insurance pricing. It quantifies a mediation channel from climate exposure to premiums and onward to financial distress, and expresses effects in policy-ready parameters that inform debates on risk adjustment, reinsurance design, and targeted, event-triggered consumer protections.

Chapter 2: Climate Exposure and Hospital Finances (Supply Side)

High-level summary. Chapter 2 examines how the same hazards affect hospital operating margins, uncompensated care, and financing conditions (e.g., credit spreads and liquidity). Hospital-year financials are merged to local hazards and ancillary stressors (such as power outages), and analyzed using fixed-effects event studies with heterogeneity by safety-net status, market power, Medicaid expansion, and community vulnerability.

Contribution to the literature. This chapter delivers causal evidence on the financial incidence of climate shocks on providers, complementing clinical and operational literatures with balance-sheet outcomes. It clarifies mechanisms (case-mix shifts, surge strain, input-cost spikes), documents how financial frictions amplify climatic stress, and yields parameters relevant for payment reform and contingent fiscal backstops under climate risk.

Chapter 3: A Structural Policy Model of Climate-Resilient Health Financing

High-level summary. Chapter 3 develops a structural framework nesting consumers (plan choice and delinquency risk), insurers (premium setting under risk adjustment/reinsurance), and hospitals (capacity and financing constraints) under a distribution of local climate shocks. Calibrated to reduced-form elasticities from Chapters 1–2 (with a worked calibration that can focus on California), the model simulates counterfactual policies: climate-risk adjusters in payment, geographically targeted premium support, disaster-triggered cost-sharing waivers, and provider stabilization instruments.

Contribution to the literature. Methodologically, the chapter links quasi-experimental estimates to a tractable equilibrium model spanning demand and supply, enabling welfare and incidence analysis of adaptation policies. Substantively, it provides a transparent policy calculator that quantifies resilience trade-offs—affordability for households, solvency for hospitals, and equity across places and income groups—as climate extremes intensify.

Overall Contribution

Collectively, the dissertation (i) establishes policy-relevant elasticities that map climate shocks to insurance prices, household financial risk, and provider solvency; (ii) demonstrates common transmission channels across market sides using harmonized exposure definitions; (iii) advances identification by aligning hazard timing with pricing, billing, and financial-reporting calendars; and (iv) delivers a structural platform that translates reduced-form evidence into counterfactual guidance for regulators and payers. It thereby extends the climate–health economics literature by jointly characterizing demand, insurer, and provider

responses and by converting those relationships into actionable design parameters for climate-resilient health financing.

Paper 1: Financial Impacts on Patients

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Introduction

Rising temperatures, longer heat seasons, and increasingly severe wildfire smoke are changing when and how Americans get sick. A large clinical and epidemiological literature shows that extreme heat days raise emergency department (ED) visits for heat illness, renal disease, and mental health conditions across the age distribution, with particularly sharp surges during acute heat events such as the 2021 Pacific Northwest “heat dome” (Kovats and Hajat 2008; Wang et al. 2021; Petkova et al. 2017; Dominici et al. 2023). Wildfire smoke adds another channel: short-term spikes in smoke $\text{PM}_{2.5}$ elevate cause-specific ED visits, especially for respiratory conditions (Vega et al. 2025; Heft-Neal et al. 2018). These morbidity shocks are now frequent enough to matter for health financing, not only for one-off disaster response but—crucially—for how routine coverage is priced and for whether households can keep up with medical bills.

This paper investigates two linked financial outcomes of climate shocks at population scale: (i) premiums in the Affordable Care Act (ACA) individual market and (ii) the prevalence and size of medical debt reported to credit bureaus. Premiums are the primary market-wide price signal of expected claims risk for non-group coverage; when expected morbidity rises, actuaries incorporate those costs (and the uncertainty around them) into rates, subject to risk adjustment and market rules (Kautter et al. 2014; American Academy of Actuaries 2025a). Medical debt, by contrast, captures household-level financial distress when cost sharing, denials, or uncovered care exceed liquid resources. Bringing these outcomes together allows us to trace how climate-driven morbidity can translate into both higher community-level premiums and greater household financial strain.

We build on and connect two literatures that have largely developed in parallel. The first documents health utilization effects of heat and smoke, including multi-state and national analyses showing higher ED visits for all-cause and cause-specific conditions on extreme heat

days and during smoke events (Bobb et al. 2014; Anderson and Bell 2009; Wang et al. 2021; Vega et al. 2025). The second quantifies the scale and geography of medical debt and its policy sensitivity. Using consumer credit reports, Kluender et al. (2021) estimated that medical debt in collections was the most common form of debt in collections and disproportionately concentrated in lower-income counties, especially in the South. Recent descriptive updates from the Urban Institute show sharp declines in reported medical collections after credit-reporting reforms in 2022–2023, but tens of millions of adults still have medical debt on their files (Urban Institute 2024). Our contribution is to unify these strands by asking whether—and how quickly—local climate shocks feed through to higher benchmark premiums and to increases in medical debt.

On the pricing side, ACA premiums provide a clean, policy-relevant measure. Benchmark premiums (second-lowest-cost silver plan, SLCSPP) are observed annually by rating area and summarize the competitive cost of covering a standard risk profile. Actuarial guidance and federal documentation make clear that rates depend on expected claims trends, benefit design, issuer competition, and the risk adjustment and high-cost risk pool that transfer funds from plans with lower-risk enrollees to those with higher-risk enrollees (HHS 2021; CMS 2025; American Academy of Actuaries 2024, 2025b). If climate shocks reliably raise expected utilization and severity for heat- and smoke-sensitive conditions, premiums could adjust upward in places where such shocks are becoming more common, all else equal. Observed county-level premium growth in recent years underscores that local factors matter (KFF 2024).

On the household side, credit-report–based measures of medical debt offer national coverage and county granularity over the past decade. They have become the standard metric for tracking medical collections, including in peer-reviewed work (Kluender et al. 2021). However, measurement changed materially when the three nationwide credit bureaus removed paid medical collections, extended the reporting window to one year, and—starting April 2023—deleted medical collections under \$500. In 2025, the Consumer Financial Protection Bureau (CFPB) finalized a rule to remove medical bills from credit reports used by lenders, further signaling ongoing policy evolution (CFPB 2025). Analyses must therefore handle these break points carefully; we do so with difference-in-differences and difference-in-difference-in-differences designs that benchmark medical collections against non-medical collections before and after the reforms (see Methods).

Our empirical strategy proceeds in two steps. First, we estimate whether county-aggregated climate shock measures (lagged to match insurer pricing cycles) predict higher SLCSPP premiums in the corresponding rating areas. We draw plan-level data from CMS Exchange Public Use Files and HIX Compare and construct county-level premium indices using rating-

area crosswalks and population weights (CMS 2025; Urban Institute 2025). Second, we ask whether the same shocks predict higher county-year medical debt in collections—and whether premium increases mediate part of that effect. This two-equation approach lets us speak to both the supply-side pricing response and the downstream financial exposure of households.

The exposures we study reflect the primary, policy-relevant climate hazards that plausibly affect near-term morbidity and costs: (i) extreme heat days defined relative to local historical distributions, derived from NOAA station-level daily data; (ii) wildfire smoke intensity using daily $\text{PM}_{2.5}$ from EPA’s AQS, with smoke-day indicators; and (iii) federally declared disasters that capture floods and severe storms. These measures map naturally to clinical mechanisms documented in prior work—dehydration, renal stress, cardiovascular strain, respiratory exacerbations—and to operational burdens observed by health systems during major events. We collapse daily exposures to county-year metrics and emphasize leads/lags appropriate to premium setting (lagged a year) and to billing/collections cycles for debt (several-month to annual lags).

Why is it important to study premiums and debt jointly? First, they capture complementary incidence. Premiums spread expected climate-driven costs across the insured pool in a rating area; medical debt reflects residual, highly regressive burdens when cost sharing and uncovered care meet low liquidity or gaps in coverage. Second, premiums can themselves affect debt: higher gross premiums (net of subsidies) may push some households to select skimpier coverage, delay payment, or fall behind on other bills, increasing arrears risk. Third, both outcomes are central to policy: state-based reinsurance programs and risk adjustment aim to stabilize premiums, while federal and state credit-reporting changes aim to mitigate the collateral damage of medical billing on households’ access to credit. Quantifying climate’s contribution to each outcome helps regulators and payers target adaptation levers—e.g., climate risk adjusters in payment, zero cost sharing for heat-sensitive conditions on extreme days, or strengthened disaster grace periods in billing.

We advance the literature in three ways. Conceptually, we articulate a clear causal chain from localized climate shocks to market-wide premiums and household medical debt, integrating clinical, actuarial, and consumer-finance perspectives. Empirically, we assemble a national county–rating-area panel (2014–2025 for premiums; 2011–2023 for debt) that links CMS/HIX premium data to Urban Institute’s county-year medical debt series, while controlling for insurer competition, state reinsurance, and shifts in credit-reporting policy. Methodologically, we deploy a stacked event-study and a triple-difference strategy that exploit (a) the timing of the 2022–2023 credit-reporting reforms and (b) cross-county contrasts in shock intensity, with non-medical collections as a counterfactual outcome. Together, these choices allow us to isolate climate-related variation from confounders and to interpret

magnitudes on a policy scale (e.g., dollars per member per month; percentage-point changes in the share with medical debt in collections).

Finally, this question is timely. Benchmark premiums rose again in 2024 and 2025 in many counties—even before subsidies are accounted for—and medical debt remains widespread despite recent reporting reforms (KFF 2024; Urban Institute 2024). As climate extremes accelerate, regulators, payers, and providers need evidence on whether and how climate risk is already being priced into health insurance and pushed onto households. By estimating the premium and medical-debt responses to heat and smoke—and by testing whether premium changes mediate debt—we provide actionable parameters for evaluating adaptation policies in the health sector.

1 Data and Methods

This paper asks how local environmental shocks are priced into health insurance and how they show up in household financial stress. We treat the environment as the source of quasi-exogenous variation and study two outcomes in parallel: (i) benchmark premiums in the ACA individual market and (ii) the prevalence and amounts of medical debt in collections recorded on consumer credit files. The empirical strategy builds two panels aligned to the institutional timing of pricing and billing: a rating-area-by-year panel for premiums and a county-by-year panel for medical debt. We then relate both outcomes to lagged measures of extreme heat, wildfire smoke/particulate pollution, and federally declared disasters. Because premiums can plausibly serve as a pathway from shocks to debt, we also provide an exploratory mediation decomposition, but all primary estimates keep premiums out of the debt equation to avoid post-treatment conditioning (Imai, Keele, and Tingley 2010). The following maps from the Climate Vulnerability Index illustrate the locations within the U.S. where climate exposure and mortality coincide (Environmental Defense Fund 2025).

Figure 1: Climate Impacts by County - Hot and Cold Days

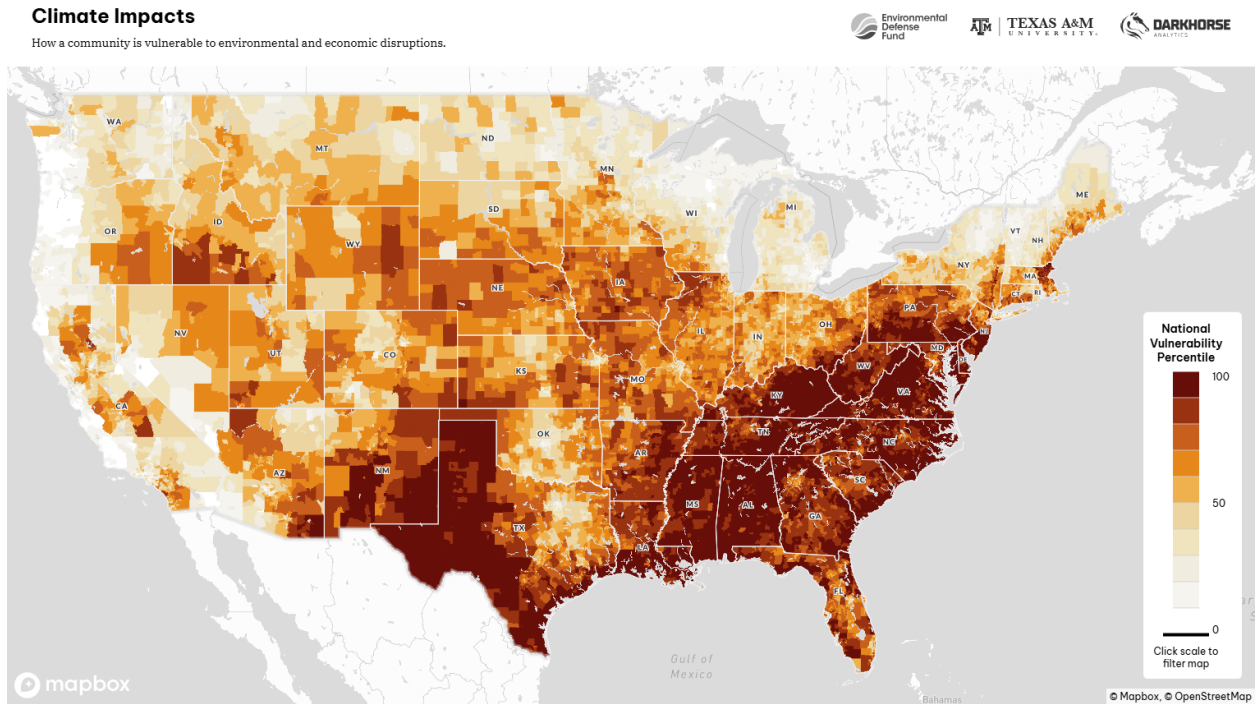
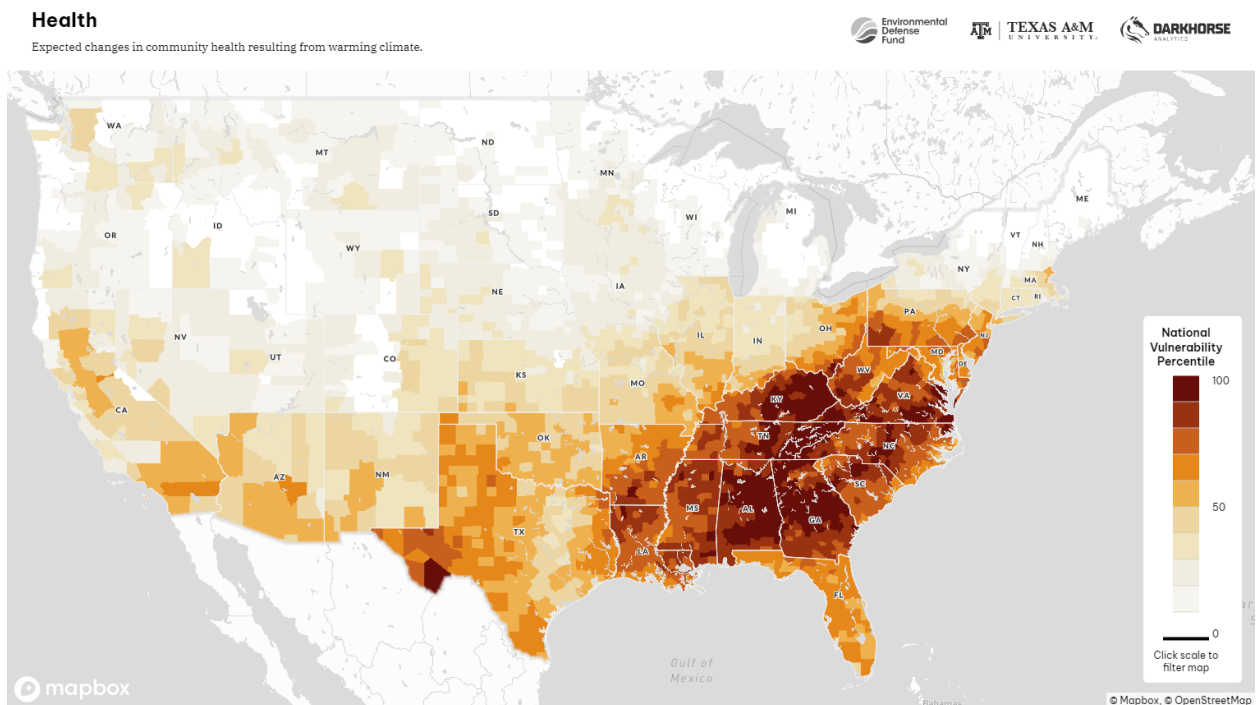


Figure 2: Health Impacts (Mortality) by County - Hot and Cold Days



1.1 Outcomes and Measurement

Medical debt. We measure household financial distress using the Urban Institute’s county–year series based on a large, nationally representative consumer credit panel. For each county and year, the dataset reports the share of adults with any *medical debt in collections* and the *median balance* among those with medical debt. The reference month is August; suppressed cells with small samples are excluded by the provider, so our panel is unbalanced by design. Credit-reporting reforms in mid-2022 and 2023—removal of collections less than 12 months old and deletion of medical collections under \$500—created visible, mechanical breaks in measured prevalence. Several states subsequently enacted prohibitions on reporting medical debt to credit bureaus. We therefore include rich time and policy fixed effects, and we verify that our conclusions are robust when truncating the series at 2021 and when excluding states affected by bans (Kluender et al. 2021; Urban Institute 2024).

Table 1: County-Level Medical Debt, Summary Statistics by Year (Urban Institute County File, 2011–2023)

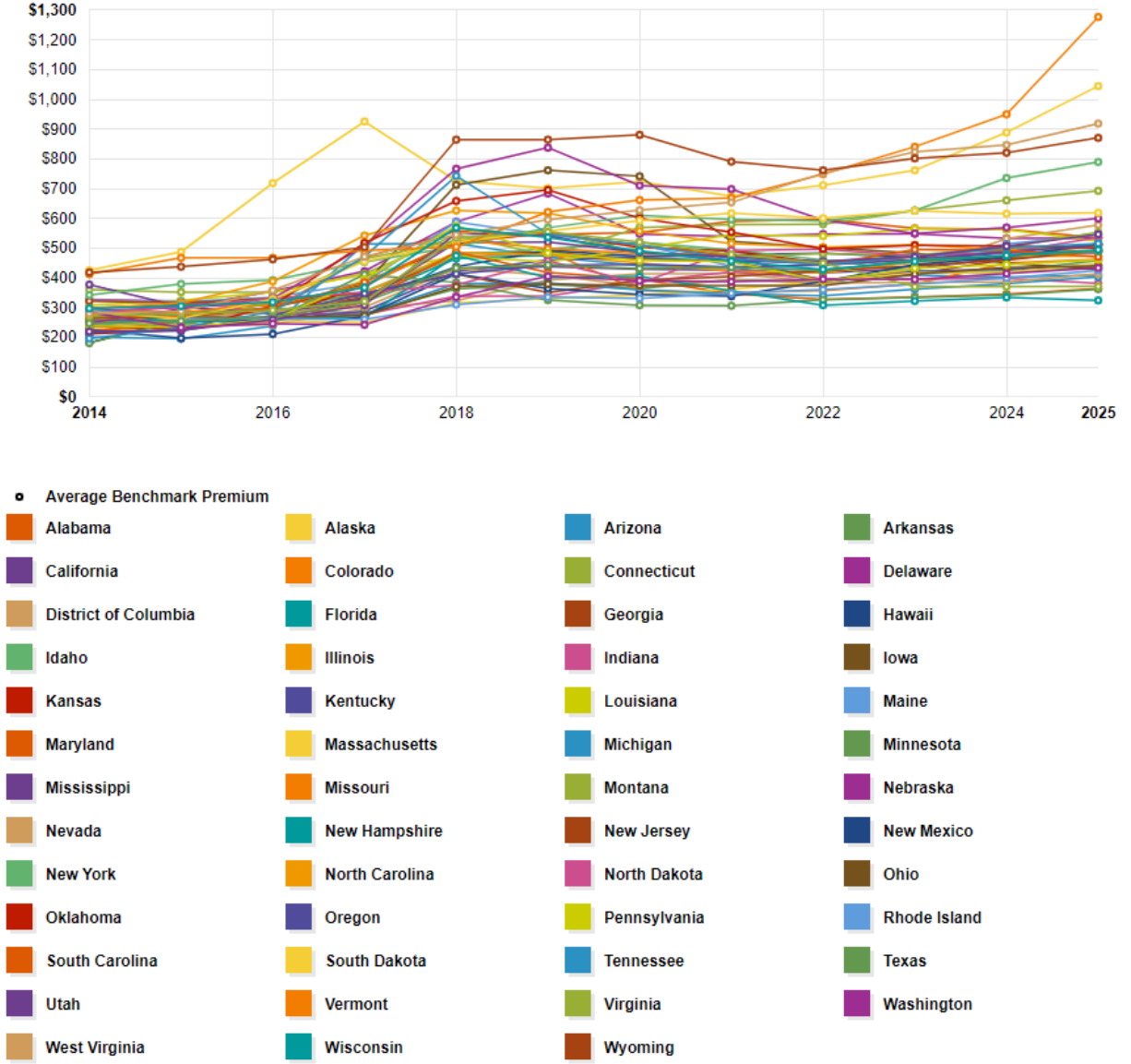
Panel A: Share of adults with medical debt in collections (%)										
Year	N (counties)	Mean	SD	P10	P25	Median	P75	P90	Min	Max
2011	2,967	21.6	9.6	8.9	14.9	21.4	28.3	34.2	0.0	51.8
2012	2,964	21.9	9.5	9.2	15.3	22.0	28.2	34.1	0.0	54.4
2013	2,967	22.7	9.9	9.5	15.8	22.7	29.5	35.3	0.0	54.4
2014	2,963	22.4	9.8	9.3	15.6	22.4	29.3	35.0	0.0	51.5
2015	2,961	22.0	9.8	8.7	15.2	22.0	29.0	34.7	0.0	51.3
2016	2,958	21.1	9.6	8.3	14.3	20.9	27.8	33.6	0.0	51.8
2017	2,947	20.0	9.4	7.7	13.2	19.7	26.7	32.6	0.0	51.1
2018	2,945	19.8	9.4	7.5	13.0	19.2	26.0	32.5	0.0	53.4
2019	2,940	18.6	9.0	6.8	12.5	18.3	24.4	30.6	0.0	56.0
2020	3,063	17.6	8.8	6.0	11.3	17.2	23.2	28.9	0.0	60.0
2021	3,067	16.8	8.5	5.4	10.6	16.6	22.6	28.2	0.0	50.5
2022	3,065	14.3	7.6	4.1	8.5	13.9	19.5	24.5	0.0	43.5
2023	3,063	6.3	4.4	1.3	3.1	5.6	8.6	12.2	0.0	32.5

Panel B: Median medical debt in collections (2023 \$)										
Year	N (counties)	Mean	SD	P10	P25	Median	P75	P90	Min	Max
2011	2,063	904	329	550	676	848	1,068	1,318	245	3,683
2012	2,071	901	333	533	670	851	1,082	1,322	198	2,690
2013	2,108	925	346	537	680	876	1,094	1,366	210	3,263
2014	2,087	933	351	548	685	889	1,106	1,364	266	3,870
2015	2,077	929	354	521	686	888	1,119	1,358	193	3,733
2016	2,030	929	351	525	675	884	1,125	1,366	225	3,278
2017	1,972	906	357	505	649	853	1,091	1,340	250	3,022
2018	1,949	912	365	492	636	855	1,130	1,377	206	3,128
2019	1,873	1,004	385	561	707	947	1,246	1,502	286	3,111
2020	2,352	979	401	516	677	915	1,221	1,513	194	2,950
2021	2,318	936	370	507	653	882	1,167	1,420	223	3,028
2022	2,203	783	318	430	542	730	960	1,211	222	2,733
2023	1,515	1,534	396	1,124	1,285	1,469	1,694	2,001	792	4,500

Notes: Each row summarizes county-level values for the indicated year. In Panel A, entries are in percentage points of adults with a credit record who have any medical debt in collections. In Panel B, entries are dollar amounts (2023 \$) of the median medical debt in collections among adults with any medical debt in collections. Statistics are unweighted across counties; N reports the number of counties with non-missing data for the metric in that year. Policy/reporting changes in credit bureaus in July 2022 (removal of medical collections under one year) and April 2023 (removal of medical collections under \$500), as well as Colorado’s removal of medical debt from credit reports in Aug 2023, may affect year-over-year comparability.

Premiums. We proxy market-wide expected claims costs with the *second-lowest cost silver plan (SLCSP)* premium for a 40-year-old, non-tobacco individual in the ACA individual market. Using CMS Exchange Public Use Files, we identify silver qualified health plans within each *rating area-year*, rank by the relevant individual premium, and select the SLCSP following CMS rules for county availability. Because rating areas are the pricing unit, our premium regressions are estimated at the rating-area level. For maps and subgroup analyses, we map rating areas to counties with the Urban Institute/RWJF HIX Compare crosswalks and compute population-weighted county SLCSPs. To characterize market context, we supplement with issuer participation (distinct QHP issuers by geography) and state-year indicators of section 1332 reinsurance programs, including simple measures of program generosity drawn from waiver documents and federal summaries. These variables serve as controls that soak up large, policy-driven premium movements unrelated to climate (CMS 2025; CCHIO 2024).

Figure 3: Premiums by State - 2014 to 2025



1.2 Environmental Exposures

The key regressors are county-level measures of hazards that plausibly increase short-run morbidity and claims costs: *extreme heat*, *wildfire smoke/PM_{2.5}*, and *federally declared disasters*. We build these from daily or incident-level administrative sources.

1. **Heat.** From NOAA's GHCN-Daily, we compute two complementary metrics for each county: (a) the *count of hot days* defined relative to local climate—days with $T_{\max}$ above the county-specific 95th percentile of the 1991–2020 distribution—and (b) *heat degree-days* above 90°F. Counties with multiple stations are aggregated using population

weights, and results are robust to using gridded alternatives.

2. **Smoke/PM_{2.5}.** From EPA’s AQS monitors, we compute the *number of days with PM_{2.5} $\geq 35 \mu\text{g}/\text{m}^3$* and the *warm-season mean PM_{2.5}* (May–October). Where a county lacks a monitor, we impute from the nearest monitor within 50 km using inverse-distance weights and flag these observations; sensitivity checks exclude imputed counties to guard against measurement error.
3. **Disasters.** From FEMA’s Disaster Declarations Summaries, we code monthly county-level declarations by incident type (wildfire, flood, severe storm, hurricane) and aggregate to *counts of disaster months* per year; we also construct an *any-disaster* indicator. In robustness, we add a coarse measure of *major electric power disturbances* from DOE OE-417 reports to capture compounding heat–outage events, noting that coverage for outages is less systematic than for the other hazards.

All exposure variables are demeaned within county to align with fixed-effects identification and are winsorized at the 99th percentile to limit the influence of extreme outliers. We standardize units in tables so that coefficients can be read as effects of a within-county one-standard-deviation change in exposures.

1.3 Time Alignment and Panel Construction

Insurers set prices prospectively based on prior-year experience and forecasts, while medical debt reflects accrual, billing, and the collections pipeline. We therefore align shocks differently across the two panels. For *premiums*, exposure year $t-1$ is linked to plan year t , consistent with the typical filing and finalization calendar. For *debt*, the Urban Institute outcome reflects the stock of collections as of August in year t , so we define the exposure window as the *trailing 12 months* (September $t-1$ through August t). In practice, our baseline uses calendar $t-1$ exposures for both panels and then verifies with finer lead–lag windows in event-study plots.

The premium panel is constructed at the *rating-area-by-year* level. The debt panel is *county-by-year*. For displays and heterogeneity analyses that combine information across panels, we map rating areas to counties and compute population-weighted county SLCSPs. Our main regressions are *unweighted*, which is standard for fixed-effects identification relying on within-unit changes; as a check, we report population-weighted estimates.

1.4 Econometric Framework

We estimate two exposure–outcome relationships in parallel.

Pricing response. For rating area r in year t ,

$$P_{rt} = \beta_P^\top \mathbf{E}_{r,t-1} + \Gamma^\top \mathbf{Z}_{rt} + \alpha_r + \delta_{s(t)} + \varepsilon_{rt}. \quad (1)$$

Here, P_{rt} is the SLCSF; $\mathbf{E}_{r,t-1}$ are lagged environmental exposures aggregated from counties within the rating area (population-weighted); $\mathbf{Z}_{rt}$ includes issuer participation and state-year reinsurance indicators/generosity; α_r are rating-area fixed effects; and $\delta_{s(t)}$ are *state* $\times$ *year* fixed effects capturing policy cycles and macro conditions. Standard errors are clustered at the state level. This specification recovers β_P , the reduced-form pricing elasticities with respect to heat, smoke, and disasters, net of broad policy shifts.

Household financial stress. For county c in year t ,

$$D_{ct} = \beta_D^\top \mathbf{E}_{c,t-1} + \Theta^\top \mathbf{X}_{ct} + \mu_c + \lambda_t + u_{ct}. \quad (2)$$

D_{ct} is either the share of adults with medical collections or the median balance among those with medical debt; $\mathbf{E}_{c,t-1}$ are lagged environmental exposures; $\mathbf{X}_{ct}$ contains local macro controls (unemployment, wages) and issuer participation to capture evolving health-system dynamics; μ_c are county fixed effects; and λ_t are year fixed effects that also absorb the national credit-reporting reforms of 2022–2023. We additionally include *state* $\times$ *post* indicators for states that adopted bans on reporting medical debt. Standard errors are clustered at the state level and, in sensitivity, with spatial HAC corrections.

Two design features help validate identification. First, we present *stacked event studies* around “shock years” (e.g., top-decile increases in heat or smoke) using estimators that are robust to staggered timing and heterogeneous treatment effects, and we explicitly show pre-trend coefficients near zero (Sun and Abraham 2021; Callaway and Sant’Anna 2021). Second, to guard against the possibility that measurement changes in credit reporting contaminate medical-debt dynamics, we re-estimate using *non-medical collections* as a placebo outcome and estimate a *triple-difference* that compares medical vs. non-medical collections, high- vs. low-shock counties, and pre- vs. post-reform periods.

1.5 Mediation Analysis (Supplementary)

Premiums may be one pathway through which environmental shocks raise medical debt—e.g., via changes in net prices for unsubsidized consumers, plan menus, or network breadth. To quantify this channel without biasing the main estimates, we conduct a simple *product-of-coefficients* decomposition outside the baseline model. We first estimate the exposure $\rightarrow$ premium

relationship to obtain $\hat{\beta}_P$. We then estimate an *augmented* debt model that includes contemporaneous premiums and obtain $\hat{\gamma}$, the association between premiums and debt conditional on exposures and fixed effects. The implied indirect effect is $\hat{\beta}_P \times \hat{\gamma}$. Because this approach relies on sequential ignorability and rules out interactions between the exposure and mediator, we present it as *exploratory* and report standard sensitivity analyses following Imai, Keele, and Tingley (2010).

1.6 Controls, Scaling, and Interpretation

Local macroeconomic conditions—unemployment and wages—enter the debt equation to improve precision and remove obvious cyclical confounds. Issuer participation and reinsurance policies enter the premium equation by construction and the debt equation only as contextual controls. We standardize continuous exposures and controls (z-scores) to aid interpretation across hazards; coefficients in tables can thus be read as the effect of a one-standard-deviation increase in heat, smoke, or disasters on premiums (in dollars per month) and on debt (in percentage points of adults with medical collections or dollars of median balances).

Because premium list prices are only imperfectly informative about *net premiums* faced by subsidy-eligible enrollees, we treat the APTC environment as context rather than an exposure. In heterogeneity analyses, we stratify counties by the share of on-exchange enrollees receiving APTC and by issuer competition (terciles of issuer counts) to assess whether pricing responses and debt impacts are larger where consumers are more exposed to gross prices or where markets are less competitive.

1.7 Robustness and Design Diagnostics

We probe the stability of results in several directions. First, we vary *exposure definitions*—absolute vs. percentile heat thresholds ($\geq 90^\circ\text{F}$, $\geq 95^\circ\text{F}$), PM_{2.5} thresholds (25/30/35 $\mu\text{g}/\text{m}^3$), and season-specific aggregations (JJA for heat; May–October for smoke)—and confirm that patterns are not artifacts of a single cutoff. Second, we add *leads* of exposures in both equations; premiums should not respond to contemporaneous-year shocks if pricing uses prior information. Third, we address policy confounding: for premiums, we retain reinsurance controls and verify that excluding adoption years does not overturn the results; for medical debt, we truncate at 2011–2021 and exclude states with post-2023 reporting bans. Fourth, we implement *placebo outcomes* (non-medical collections for the debt equation; small-group premiums, where available, for the premium equation) to test whether effects are specific to the settings we theorize. Finally, we examine *distributional heterogeneity* using CDC’s Social Vulnerability Index and baseline uninsured rates; if climate-linked financial stress is

regressive, effects should be larger in high-SVI, high-uninsurance counties.

1.8 Limitations

Two measurement issues deserve emphasis. Medical-debt outcomes come from credit reports and thus reflect both underlying liabilities and the *rules by which those liabilities are reported*. The national reporting reforms of 2022–2023 and subsequent state bans complicate time-series interpretation; our fixed effects, state-specific post indicators, truncation checks, and triple-difference placebos substantially mitigate—but cannot fully eliminate—these concerns (Kluender et al. 2021; Urban Institute 2024). Environmental exposures are measured with error: station coverage varies, some counties lack air-quality monitors, and disaster declarations are administrative constructs. We address these with transparent construction rules, imputation flags, alternative sources and thresholds, and sensitivity excluding imputed counties. Lastly, premiums are *gross list prices* for a standardized risk profile; net prices vary with income and family composition through APTC. Our interpretation is therefore explicitly about how *expected claims costs are priced into list premiums* and how *shocks relate to the stock of medical collections*, not about individual affordability per se.

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Paper 2: Financial Impacts on Providers

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Introduction

Hospitals sit at the financial frontier of climate change. When extreme heat and wildfire smoke raise acute morbidity—especially respiratory and cardiovascular exacerbations—care shifts abruptly toward emergency and inpatient services. These surges stress capacity, raise variable costs, and can erode operating margins, particularly in safety-net settings. Multiple large datasets now document these acute-care spikes. During the 2023 warm season, CDC’s National Syndromic Surveillance Program recorded substantial increases in heat-related ED visit rates across U.S. regions relative to prior years, with sustained elevations over weeks, underscoring how extreme heat converts quickly into acute service demand. (Vaidyanathan et al. 2024, MMWR.) Wildfire smoke produces similar shocks: in California, hospital-catchment-level analyses show that wildfire-specific $PM_{2.5}$ exposure is associated with higher all-cause and respiratory ED visits; complementary cohort evidence links high smoke pollution days to increased hospitalizations for respiratory disease. (Stowell et al. 2025; Liu et al. 2017; Vega et al. 2025.)

These clinical shocks are not financially neutral. One provider-side transmission mechanism runs through uncompensated care. Uninsured and underinsured patients who present during climate events can generate spikes in charity care and bad debt. In the U.S., these costs are measured on Medicare Cost Report Worksheet S-10, which standardizes charity/bad-debt reporting, anchors the distribution of disproportionate share hospital (DSH) uncompensated-care payments, and is now the focus of an ongoing federal audit regime to ensure accuracy. (CMS 2024.) If climate shocks predictably raise ED/inpatient encounters among financially vulnerable populations, S-10 totals should rise in exposed hospitals and years—pressuring cash flow precisely when capacity must be expanded to meet acute demand.

A second mechanism is simply the cost of acute episodes themselves. Heat-related illness (HRI) admissions and smoke-linked exacerbations often involve ED evaluation, respiratory

support, ICU observation, and follow-up. U.S. studies estimate meaningful inpatient costs for HRI and show that the burden falls disproportionately on patients from low-income zip codes and those without insurance—exactly the cases most likely to produce uncompensated charges. (Schmeltz et al. 2016; Schmeltz et al. 2015.) More broadly, recent evidence continues to find that wildfire-derived $PM_{2.5}$ raises risks of hospital admission and ED utilization for cardiorespiratory illness, reinforcing the expectation of higher variable costs during smoke episodes. (Wang et al. 2025.)

Crucially, both heat and smoke shocks are socially patterned, amplifying hospitals’ exposure to uncompensated care. Persons experiencing homelessness, for example, had approximately 30-fold higher rates of heat-related ED visits than housed adults in 2021-2022 and were twice as likely to be hospitalized once in the ED—concentrating costly episodes in hospitals serving unhoused populations. (Weckstein et al. 2025, JAMA Internal Medicine.) Parallel work finds that HRI hospitalizations are most prevalent in the lowest-income zip-code quartile and among the uninsured, and that heat-related emergencies cluster in socially vulnerable counties as measured by CDC’s Social Vulnerability Index (SVI). (Schmeltz et al. 2015; Lehnert et al. 2020; Derakhshan et al. 2023.) Wildfire smoke also elevates mental-health-related ED visits—particularly for children, women, and racial/ethnic minority groups—implying added demand on emergency and observation services beyond cardiorespiratory care. (Jung et al. 2025; Zhu et al. 2024.) Together, these findings point to a systematic pattern: climate shocks disproportionately drive acute episodes among populations with limited ability to pay, increasing hospitals’ uncompensated care and compressing operating margins.

Beyond operating performance, climate risk appears to raise the cost of capital for healthcare providers, tightening financial constraints exactly where operating shocks are largest. A recent Dallas Fed working paper shows that wildfire smoke pollution increases healthcare municipal borrowing costs, with roughly \$250-\$270 million in incremental interest expenses for high-smoke counties during 2010-2019, and disproportionately larger penalties in high-poverty or high-minority areas where smoke-related uncompensated care is greater. (Lopez, Murphy, Tzur-llan, and Wilkoff 2025.) Although the current paper does not study capital markets directly, that result strengthens the policy case: if climate shocks both raise current-period uncompensated care and increase future borrowing costs, then safety-net hospitals confront a two-front financial challenge—working-capital strain today and more expensive adaptation investments tomorrow.

Despite these plausible and increasingly well-documented pathways, the provider-side financial footprint of climate shocks remains incompletely measured at national scope. We still lack estimates of how extreme heat and wildfire smoke affect (i) uncompensated care (S-10) and (ii) operating performance (e.g., operating margin proxies) in a hospital-year

panel, with attention to heterogeneity by safety-net status, rurality, and state policy context. Yet the necessary accounting and administrative data exist: CMS’s Healthcare Cost Report Information System (HCRIS) provides standardized hospital financials and is regularly updated; S-10 totals are used in payment formulas and subject to audit; and hospital characteristics from the American Hospital Association (AHA) enable profiling of capacity and safety-net roles. (CMS HCRIS 2025; CMS S-10 audit page 2024.) Linking these sources to high-resolution measures of heat and smoke exposures makes it possible to move from clinical risk to financial impact at the facility level.

This paper delivers that evidence. First, we assemble a hospital-year panel merging HCRIS financials with S-10 uncompensated-care totals and AHA survey characteristics. Second, we construct local climate exposures—annualized heat intensity (e.g., high-heat degree-days) and smoke intensity (e.g., plume-day counts or wildfire-specific $PM_{2.5}$)—aggregated to hospital catchments. Third, we estimate how within-hospital variation in heat and smoke intensity affects uncompensated care and operating performance using event-study and difference-in-differences designs with high-dimensional fixed effects to absorb persistent differences in hospital type and location. Robustness includes alternative exposure definitions, placebo off-season exposures, leave-one-season-out checks, and complementary specifications that compare hospitals in adjacent geographies exposed to different smoke/heat realizations.

Our contribution is threefold. Conceptually, we frame climate change as a provider-finance problem, clarifying how acute environmental shocks translate into cash-flow strain through uncompensated care and variable costs. Empirically, we quantify national-scope effects of heat and smoke on S-10 and operating performance, with equity-relevant heterogeneity. Policy-wise, because S-10 already anchors DSH allocations and is under an expanding audit regime, our estimates can inform whether climate-sensitive adjustments—for example, targeted uncompensated-care pools during declared heat/smoke events or support for hospital-level adaptation—would better stabilize facilities serving highly exposed communities. (CMS 2024.)

In short, extreme heat and wildfire smoke predictably raise acute hospital demand, especially among populations with limited ability to pay. That combination increases uncompensated care and compresses operating performance—and, as separate capital-market evidence suggests, may also raise the price of resilience investments over time. This paper focuses squarely on the provider side to measure those operating impacts and their distribution across the U.S. hospital sector. (Vaidyanathan et al. 2024; Stowell et al. 2025; Liu et al. 2017; Jung et al. 2025; Lopez et al. 2025.)

Data and Methods

Our empirical strategy treats climate shocks as plausibly exogenous year-to-year fluctuations that differentially strike hospitals within the same state and over time. We assemble an annual hospital-year panel spanning fiscal years 2011-2023, which aligns with the period in which Worksheet S-10 uncompensated care reporting stabilized and entered payment policy and auditing more systematically (CMS 2024, 2025). The unit of observation is the hospital (h) in fiscal year (t). We focus on two provider-side outcomes that directly map to operating finances: uncompensated care costs reported on Worksheet S-10 (charity care plus non-Medicare bad debt), and operating performance constructed from Medicare Cost Reports (HCRIS) as operating income divided by operating revenue (CMS 2025). These outcomes capture, respectively, the extent to which hospitals deliver care without payment during shock years and the net operating consequences of those episodes.

To link clinical shocks to hospital finances, we combine three data systems. First, we extract uncompensated care aggregates from S-10 and financial statements from HCRIS public use files, harmonizing identifiers and fiscal years. Second, we bring in the American Hospital Association (AHA) Annual Survey to characterize ownership (public, nonprofit, for-profit), teaching status, staffed beds, system membership, and location, allowing us to describe heterogeneity in responses by safety-net status and rurality (AHA 2025). Third, we attach local exposure measures for heat and wildfire smoke. Heat is measured from NOAA’s Global Historical Climatology Network daily records as the annual count of very hot days—our baseline uses ($T_{\max} \geq 95^{\circ}\text{F}$)—with robustness to alternative thresholds and to cooling-degree-days; where humidity data are available, we compute heat-index thresholds using the National Weather Service’s Rothfusz equation (NOAA NCEI 2019+; NWS/WPC 2022). Wildfire smoke is measured in two complementary ways: the number of “smoke plume” days in which NOAA’s Hazard Mapping System identifies analyst-curated smoke polygons over a county, and a dose proxy built by summing daily $PM_{2.5}$ observed at EPA AQS monitors on plume days and averaging across monitors in the county (NOAA NESDIS 2024-25; EPA 2025). Using both presence (plume days) and dose (smoke- $PM_{2.5}$) helps triangulate the relevant channel.

Table 1: Summary statistics by year: Uninsured and Bad Debt Cost as % of Net Patient Revenue

Year	N	Mean	SD	Min	P25	Median	P75	Max
2011	4356	0.032	0.049	0.000	0.010	0.020	0.037	1.243
2012	4592	0.035	0.044	0.000	0.012	0.024	0.042	0.962
2013	4559	0.037	0.048	0.000	0.013	0.025	0.044	0.915
2014	4538	0.035	0.043	0.000	0.012	0.023	0.042	0.868
2015	4515	0.031	0.041	0.000	0.010	0.019	0.036	0.718
2016	4492	0.031	0.040	0.000	0.009	0.018	0.036	0.717
2017	4484	0.029	0.036	0.000	0.009	0.017	0.034	0.462
2018	4451	0.027	0.036	0.000	0.009	0.016	0.031	0.646
2019	4399	0.026	0.042	0.000	0.008	0.016	0.030	1.698
2020	4361	0.026	0.039	0.000	0.009	0.016	0.031	1.248
2021	4314	0.022	0.058	0.000	0.007	0.012	0.025	3.281
2022	4310	0.020	0.030	0.000	0.006	0.012	0.023	1.046
2023	4264	0.019	0.030	-0.002	0.006	0.011	0.021	0.639

Hospitals are geocoded from cost-report addresses and mapped to county FIPS codes, which serve as our baseline exposure geography. Because many hospitals draw from multi-county markets, we assess sensitivity to catchment definitions by re-assigning exposures at the Dartmouth Atlas hospital service area (HSA) level via the ZIP HSA crosswalk (Dartmouth Atlas, various years). In the main results we keep the exposure temporal resolution annual to match financial reporting; in sensitivity analyses we explore fiscal-year alignment and alternative calendar windows. Outcomes are expressed both in levels and in ratios that scale by hospital operations. For S-10 we report totals in 2023 dollars (deflated using CPI-U) and the share of uncompensated care relative to net patient revenue; for operating performance we use an operating margin proxy and, in checks, operating income per bed. We winsorize outcomes at the 99th percentile to limit the influence of reporting outliers and large, infrequent events, and we verify that results are similar without winsorization. The analytic sample includes short-term acute-care hospitals filing CMS-2552-10 with non-missing S-10 and HCRIS financials; federal and specialty hospitals are excluded. When providers merge or change CCNs, we follow CMS cross-references and AHA system metadata to maintain longitudinal continuity.

$$Y_{ht} = \beta_1 \text{Heat}_{ct} + \beta_2 \text{Smoke}_{ct} + \eta_h + \delta_{st} + \mathbf{X}_{ht}\theta + \epsilon_{ht}$$

The main specification is a standard two-way fixed-effects model with parsimonious time-varying controls (bed count, system status) and with standard errors clustered at the hospital level. We weight by staffed beds in the primary analysis to reflect fiscal scale but show unweighted estimates as a check. Because uncompensated care is non-negative and right-skewed, we verify robustness using PPML on levels in addition to OLS on levels and shares. To probe dynamics and the parallel-trends requirement, we estimate an event-study centered on "extreme exposure years": for each hospital, we define years in the top decile of its historical heat or smoke distribution and trace outcomes from three years before to three years after the event. Flat pre-trends provide a diagnostic that, absent the shock, outcomes for that hospital would have evolved similarly to its own trend.

Two low-cost design enhancements bolster credibility. First, we repeat the analysis with "border-pair" fixed effects that compare hospitals in adjacent counties on either side of county borders (and, where feasible, within the same state) while including national year effects. These comparisons tighten geographic controls and exploit the fact that smoke and heat can vary sharply over short distances due to wind and topography. Second, we include a seasonal placebo that counts "smoke" or "heat" in off-season months when true exposure is implausible; null effects there support the interpretation that our measured relationships reflect genuine climate shocks rather than generic seasonal confounding.

Given concern about measurement error in smoke exposure, we report results using both HMS plume days and smoke- $PM_{2.5}$ and include annual mean $PM_{2.5}$ as a conditioning variable so that estimated smoke effects are not simply capturing background air quality trends (EPA 2025). We further show that results are not driven by a single extraordinary season by re-estimating after dropping each year in turn and by excluding the COVID-19 years (2020-2021), during which utilization and finances were affected by extraordinary policy and behavioral responses. Because reporting incentives could, in principle, bias S-10 totals, we note that CMS has implemented an audit regime and uses multi-year averages in DSH allocation, which reduces the payoff to manipulating year-to-year numbers; empirically, our pre-trend tests and placebo seasons provide additional reassurance (CMS 2024, 2025).

Equity and differential exposure are central to the question at hand, so we examine heterogeneity along three pre-specified axes. We interact exposures with indicators for safety-net orientation (public ownership; high Medicaid/charity burden where measurable), with rurality using NCHS urban-rural county codes (or RUCA codes as an alternative), and with quartiles of the CDC/ATSDR Social Vulnerability Index (NCHS 2014; USDA ERS 2025; CDC/ATSDR 2024). These interactions ask whether the same climatic shock produces larger uncompensated-care surges or sharper margin compression in hospitals serving socially vulnerable communities or operating with thinner financial buffers. Throughout, we report both dollar effects (per one-SD increase in exposure) and semi-elasticities (percentage change relative to baseline uncompensated care or operating revenue) to put magnitudes in fiscal context.

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Paper 3: Structural Model For Subsidy Targeting

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1 Introduction

Premium subsidies are the central instrument the United States uses to make individually purchased health insurance affordable. Since 2021, enhanced premium tax credits (PTCs) have lowered net premiums for low- and middle-income marketplace enrollees and expanded eligibility above 400 percent of the federal poverty line. These enhancements were extended through plan year 2025 under the Inflation Reduction Act; their fate beyond 2025 is actively debated. In parallel, proposed 2026 rates point to sizable gross premium increases amid policy uncertainty, rising medical prices, and utilization pressure—especially salient for consumers who would face the full increase if enhanced PTCs lapse. In short, the affordability problem is both immediate (what happens if temporary enhancements end) and structural (how to design subsidies that remain effective as underlying risks evolve).¹

A growing body of evidence shows that climate-related shocks—particularly extreme heat and wildfire smoke—raise acute care utilization, shifting case mix toward emergency and inpatient care and elevating spending in exposed communities. While estimates vary by setting and population, recent reviews and multi-state analyses document increased emergency department (ED) visits and hospitalizations during hot spells and severe smoke days, including for cardiovascular, respiratory, renal, and some injury categories. These demand spikes are precisely the sorts of utilization shocks that, if persistent or recurring, enter the claims experience that underlies premium setting in

¹<https://www.kff.org/affordable-care-act/aca-marketplace-premium-payments-would-more-than-double/>

the individual market. Even if annual experience “averages out,” geographic clustering of climate shocks can amplify premium pressure in high-exposure rating areas.²

The Affordable Care Act (ACA) constrains how insurers price that risk: community rating within rating areas, age bands, and plan-metal benchmarks interact with a risk-adjustment (RA) program that transfers funds based on enrollee morbidity. RA is not designed to compensate for environmental exposure per se; it equalizes for coded health risk conditional on the enrolled population. Nor does the medical loss ratio (MLR) rule—requiring at least 80 percent of premiums be spent on claims in the individual market—directly ensure that consumers are shielded from climate-driven volatility. In practice, premiums still reflect expected claims and administrative loads, and pass-through from costs to premiums can be substantial. Understanding how RA and MLR shape pass-through is crucial for calibrating premium subsidies that maintain affordability without dulling price discipline.³

This paper argues that **premium subsidies should be calibrated to the evolving distribution of climate risk and to measured pass-through into premiums**, with safeguards to preserve incentives for efficiency and plan competition. Calibrating means choosing the *level* (how generous), the *slope* (how rapidly subsidies rise as gross premiums increase), the *targeting rule* (which people and places), and the *interaction* with payment-side stabilizers (risk adjustment, reinsurance, and any climate add-on payments). Each of these levers carries trade-offs:

1. **Level and slope.** A more generous subsidy schedule that tightly pegs net premiums to income can stabilize enrollment and protect vulnerable households when gross premiums rise. But if pass-through from subsidies to premiums is high, overly steep schedules risk blunting competitive pressure and raising federal outlays without commensurate consumer benefit. A calibrated slope requires empirical discipline: (i) the cost-to-premium pass-through rate, (ii) the income-gradient of enrollment elasticity, and (iii) how much of observed premium growth reflects transitory vs. persistent (e.g., climate-linked) drivers.⁴
2. **Targeting rule.** Geography matters. Rating areas with recurrent heat/smoke shocks may face systematically higher claims volatility even

²<https://pmc.ncbi.nlm.nih.gov/articles/PMC9623446/>

³<https://pmc.ncbi.nlm.nih.gov/articles/PMC4214269/>

⁴<https://www.healthsystemtracker.org/brief/how-much-and-why-aca-marketplace-premiums-are-going-up/>

after RA. Calibrating subsidies can recognize this by modest, rules-based adjustments tied to *exogenous* exposure indices (e.g., smoke-PM or heat-degree-day thresholds) rather than to realized medical spending alone. Properly designed, this reduces the risk of adverse selection out of high-exposure areas and supports risk pooling, while avoiding direct payment for inefficiency. Evidence on climate–utilization links justifies exploring such geographically contingent calibration.⁵

3. **Program interactions.** RA and MLR guardrails shape the equilibrium response to any subsidy change. If RA accurately compensates for morbidity differences but misses environmental exposure, then subsidies that keep consumers enrolled in high-exposure places can improve pooling *if* paired with payment-side tools (e.g., reinsurance or climate add-ons) that reduce residual volatility. Conversely, if MLR rules or market structure facilitate above-cost pricing, subsidy enhancements risk capture. Designing calibration jointly with payment-side reforms limits these risks.⁶
4. **Dynamics and adaptation.** Premiums reflect not only contemporaneous claims but also expectations about trend risk. If households invest in adaptation (e.g., cooling/air filtration) that reduces climate-sensitive morbidity, the “true” long-run subsidy needed to hold net premiums stable could be lower than one inferred from short-run spikes. Calibrating subsidies alongside targeted, evidence-based adaptation benefits (e.g., vouchers for efficient cooling for high-risk enrollees) can therefore be more fiscally efficient than subsidies alone.

These policy needs motivate the paper’s **structural–microsimulation framework**. The idea is simple: use a parsimonious model of patient care choice, insurer pricing, and provider finances in which climate shocks enter as exogenous risk; **discipline the key parameters with reduced-form moments from Papers 1–2** (utilization and spending responses to heat/smoke; pass-through of cost shocks to premiums; uncompensated-care responses); and then simulate counterfactual subsidy schedules under alternative climate scenarios. The outputs report not only federal outlays and average net premiums, but also **distributional impacts** (by income and

⁵<https://pmc.ncbi.nlm.nih.gov/articles/PMC9623446/>

⁶<https://pmc.ncbi.nlm.nih.gov/articles/PMC4214269/>

social vulnerability), plan participation, and potential spillovers to provider margins in safety-net settings.

From a policy design standpoint, three questions drive the calibration exercise:

- **How much pass-through should the subsidy anticipate?** If one percentage-point increase in gross premiums raises net consumer payments by less than one point because subsidies absorb the difference, enrollment remains stable—but public costs rise. If the pass-through to premiums from underlying claims is itself high, then the subsidy schedule’s slope becomes a key determinant of both take-up and fiscal cost. Recent actuarial and policy analyses document large proposed premium increases for 2026, underscoring why subsidy rules must be robust to cost shocks rather than tuned to a single year’s rates.⁷
- **What targeting rule best preserves pooling and equity?** Calibrating by income alone is necessary but may be insufficient where climate exposure is spatially concentrated. Transparent, rules-based geographic adjustments—anchored to publicly observable climate metrics and capped to limit gaming—can preserve affordability in high-exposure areas without undermining RA’s morbidity equalization.⁸
- **How should calibration interact with payment-side stabilizers?** If RA and reinsurance absorb most volatility, a flatter subsidy slope may suffice. If not, steeper subsidies can be justified—but should be paired with reforms that keep plan incentives aligned (e.g., tightening MLR definitions to reduce loopholes and ensure consumer value).⁹

The contribution of this paper is twofold. First, it provides **sufficient-statistics guidance** for policymakers: a mapping from a small set of empirically measurable objects—the pass-through rate, enrollment elasticities by income, and the covariance of climate exposure with claims—to the optimal level, slope, and targeting of premium subsidies. Second, it delivers **policy-ready counterfactuals** that compare status quo subsidies to

⁷<https://www.healthsystemtracker.org/brief/how-much-and-why-aca-marketplace-premiums-are-going-up/>

⁸<https://pmc.ncbi.nlm.nih.gov/articles/PMC9623446/>

⁹<https://www.cms.gov/marketplace/resources/data/medical-loss-ratio-data-systems-resources>

climate-calibrated schedules under realistic scenarios (e.g., persistence of current extreme-heat frequency vs. a moderate-increase path), reporting consumer welfare, coverage, and fiscal outcomes alongside plan participation and potential effects on safety-net provider finances.

The timing is apt. With enhanced PTCs set to expire after 2025 absent congressional action, states and federal policymakers are considering extensions or redesigns that will shape marketplace affordability and coverage through the mid-2020s and beyond. Rate filings and independent analyses already anticipate unusually large premium increases for 2026, driven by medical price growth and utilization. If subsidy design remains static while the risk environment becomes more volatile and spatially unequal, affordability gaps will widen precisely where climate exposure is highest. By calibrating subsidies to the empirical structure of pass-through and to transparent measures of climate risk—and by coordinating with payment-side stabilizers—policymakers can better protect coverage, sustain risk pooling, and target public dollars where they deliver the greatest welfare gains.¹⁰

Finally, this paper situates premium-subsidy calibration within a broader policy package. Subsidies, while essential, are not a substitute for investments that reduce climate-sensitive morbidity (e.g., targeted adaptation benefits) or for prudent stabilization on the payment side (risk adjustment, reinsurance, or climate add-ons). The modeling framework therefore evaluates subsidy calibration both alone and in combination with these complementary tools. The goal is not merely to cushion households against higher posted premiums, but to **design a coherent affordability architecture** that is resilient to a warming, smokier climate—and that preserves the competitive incentives at the heart of the marketplace.

2 Methods

2.1 Overview and Unit of Analysis

We implement a semi-structural, cell-based microsimulation that combines a small set of empirically estimated “adequate statistics” with a transparent accounting model of premiums, subsidies, and enrollment. The unit of analysis is a rating area $r \times$ year $t \times$ income bin y cell (optionally crossed with insurer i for pass-through estimation). Climate exposure enters as an

¹⁰<https://www.kff.org/affordable-care-act/aca-marketplace-premium-payments-would-more-than-double/>

exogenous, measured shock z_{rt} (e.g., degree-days above a local threshold or smoke-PM indicators). The sufficient statistics are:

1. *Claims sensitivity to exposure* $\beta_z = \partial C_{rt} / \partial z_{rt}$,
2. *Premium pass-through* $\rho = \partial P_{rt} / \partial C_{rt}$,
3. *Enrollment elasticity to net premium* $\varepsilon_{\text{enroll}}(y)$ by income,
4. (Optional) *Utilization/spending elasticity to out-of-pocket price* $\varepsilon_{\text{util}}$ by care setting.

Statistics (1)–(3) are estimated from reduced-form analyses; the microsimulation then evaluates policy counterfactuals by altering the *level* and *slope* of premium subsidies and, if desired, adding a rules-based geographic “kicker” tied to exposure.

2.2 Data Construction

We assemble an annual panel of rating-area cells with: (a) benchmark (silver) premiums P_{rt} and baseline enrollment by income bins E_{iyr0} ; (b) claims (or allowed amounts) C_{rt} at the rating-area or insurer–rating-area level; (c) climate exposure z_{rt} ; (d) time-varying controls X_{rt} (e.g., input cost indices, wages, demographics); and (e) crosswalks for income bins to subsidy rules. When cell-level claims are unavailable, benchmark premiums—conditional on fixed effects—serve as sufficient statistics for expected claims; pass-through is still identified by instrumenting with exposure.

2.3 Estimating the Sufficient Statistics

(1) **Claims sensitivity to exposure (β_z).** We estimate

$$C_{rt} = c_r + \lambda_t + \beta_z z_{rt} + X_{rt}\gamma + u_{rt}, \quad (1)$$

with rating-area fixed effects c_r and year effects λ_t . To capture timing and persistence, we optionally estimate a distributed-lag specification $\sum_{\ell=-L}^L \beta_\ell z_{r,t+\ell}$. Estimation is by OLS with standard errors clustered at the rating area. We mitigate exposure measurement error by (i) population-weighted tract aggregation to rating areas, (ii) excluding monitors with systematic missingness, and (iii) rich seasonality controls.

(2) Premium pass-through (ρ). At the insurer $\times$ area level:

$$P_{irt} = \alpha_{ir} + \delta_t + \rho C_{rt} + \theta (HHI_{rt} \times C_{rt}) + v_{irt}, \quad (2)$$

where α_{ir} are insurer $\times$ area fixed effects, δ_t year effects, and HHI_{rt} is market concentration. Because C_{rt} and P_{irt} may be jointly determined, we instrument C_{rt} with exposure z_{rt} and its lags using the first stage from (1), delivering a 2SLS estimate of ρ (and θ for heterogeneity by concentration). Robustness checks restrict to benchmark issuers and use within-issuer changes across adjacent rating areas.

(3) Enrollment elasticity to net premium ($\varepsilon_{\text{enroll}}(y)$). We estimate enrollment responses within income bins using either (i) a log-log panel with fixed effects,

$$\ln E_{iyrt} = \mu_{iyr} + \tau_t + \varepsilon_{\text{enroll}}(y) \ln \text{NetPrem}_{iyrt} + \eta_{iyrt}, \quad (3)$$

or (ii) a micro-logit in which the probability to enroll depends on $\ln(\text{NetPrem})$, instrumented with exposure-driven premium variation. We stratify by income, and validate elasticities around policy discontinuities in the PTC schedule.

(4) Utilization elasticity ($\varepsilon_{\text{util}}$) (optional). From event-level analyses, we recover price elasticities for utilization rates by setting (ambulatory, ED, inpatient) or by diagnosis group. When not used, claims changes are treated as exposure-driven via β_z .

2.4 Policy Model and Microsimulation

Pricing and subsidy mechanics. Given $\hat{\beta}_z$ and $\hat{\rho}$, we construct climate-driven claims and premiums:

$$\hat{C}_{rt} = \hat{c}_r + \hat{\lambda}_t + \hat{\beta}_z z_{rt} + X_{rt} \hat{\gamma}, \quad \hat{P}_{rt} = \bar{\alpha}_r + \bar{\delta}_t + \hat{\rho} \hat{C}_{rt}, \quad (4)$$

where $\bar{\alpha}_r, \bar{\delta}_t$ are set to match baseline levels. The premium tax credit (PTC) under policy $\mathcal{S}$ is parameterized as

$$PTC_{iyrt}(\mathcal{S}) = b(y) + \sigma \hat{P}_{rt} + g(z_{rt}), \quad (5)$$

where $b(y)$ is base generosity by income, $\sigma \in [0, 1]$ is the subsidy *slope* (share of premiums absorbed), and $g(\cdot)$ is an optional, capped geographic supplement for high-exposure periods. Net premium is

$$\text{NetPrem}_{iyrt} = \max\{\hat{P}_{rt} - PTC_{iyrt}(\mathcal{S}), 0\}. \quad (6)$$

Enrollment response. We update enrollment with a constant-elasticity mapping:

$$E_{iyrt}(\mathcal{S}) = E_{iyr0} \left(\frac{\text{NetPrem}_{iyrt}(\mathcal{S})}{\text{NetPrem}_{iyr0}} \right)^{\varepsilon_{\text{enroll}}(y)}. \quad (7)$$

As a robustness check, we also implement a logit version calibrated to match the same elasticity at the baseline point.

Optional utilization/OOP margin. If activated, event rates by setting $q \in \{\text{amb}, \text{ED}, \text{IP}\}$ adjust to out-of-pocket prices with elasticity $\varepsilon_{\text{util}}^q$. Claims are then the sum of unit costs times event rates, allowing substitution from ED/IP toward ambulatory care under copay waivers on shock days.

Outcomes. For each policy $\mathcal{S}$ and year t , we compute: (i) coverage (total enrollment and take-up by income/SVI tercile); (ii) affordability (median net premium and net-premium-to-income ratios); (iii) federal outlays

$$\text{Outlays}(\mathcal{S}) = \sum_{r,y} E_{iyrt}(\mathcal{S}) \times PTC_{iyrt}(\mathcal{S}); \quad (8)$$

(iv) benchmark premiums $\hat{P}_{rt}$ and their dispersion; and (v) (optional) provider pressure via a linear mapping from exposure to uncompensated care, $UC_{rt} = \kappa_0 + \kappa_z z_{rt}$.

Welfare accounting. With the logit representation of enrollment choice, the consumer-surplus change per cell is

$$\Delta CS_{iyrt} = \frac{1}{|\beta_y|} [\ln(1 + e^{V_{iyrt}(\mathcal{S})}) - \ln(1 + e^{V_{iyrt}(\mathcal{S}_0)})], \quad (9)$$

where V_{iyrt} is the deterministic utility and $\beta_y < 0$ is the coefficient on $\ln(\text{NetPrem})$. We report total welfare as $\sum \Delta CS - \Delta \text{Outlays}$ and provide distributional decompositions by income and SVI. As a sufficient-statistics check, we also report Harberger-triangle approximations using $\varepsilon_{\text{enroll}}(y)$ around baseline.

2.5 Identification and Threats to Validity

We address three concerns. *(i) Exposure validity.* We restrict to within-area variation with rich seasonality controls; we show exposure is uncorrelated with pre-trends in premiums/enrollment (placebo years) and orthogonal to input-cost shocks. *(ii) Price endogeneity.* Pass-through is estimated by IV, instrumenting claims with exposure; conditional on fixed effects and controls, exposure acts as a cost shifter with a strong first stage. *(iii) Composition effects.* As exposure may induce migration or plan exit, we reweight cells to a fixed population and replicate estimations using within-issuer changes and adjacent-area contrasts. Standard errors are clustered by rating area (or two-way clustered by area and insurer).

2.6 Calibration, Validation, and Uncertainty

Calibration. We pick $b(y)$ and σ under the status-quo policy to match observed net premiums and federal outlays in the latest pre-policy year. *Validation.* The model reproduces (i) baseline enrollment by income, (ii) cross-area variance in premiums, and (iii) elasticity tests around historical subsidy changes. *Uncertainty.* Parameter uncertainty is propagated with a block bootstrap over rating areas (resampling claims and pass-through estimations jointly) and recomputing outcomes; percentile confidence intervals are reported. Sensitivity analyses sweep ρ and β_z by $\pm 25\%$ and vary $\varepsilon_{\text{enroll}}(y)$ within literature ranges.

2.7 Policy Counterfactuals

We examine: (1) *Slope changes* $\sigma \in \{0.3, 0.5, 0.7\}$ holding $b(y)$ fixed; (2) *Level changes* increasing $b(y)$ at targeted incomes; (3) a *geographic kicker* $g(z_{rt}) = \min\{\kappa \cdot (z_{rt} - z_r^{90}), \bar{g}\} \mathbf{1}\{z_{rt} > z_r^{90}\}$; and (4) *combined packages* pairing a higher σ with a capped $g(\cdot)$. For each scenario we report national totals and cross-area distributions, plus efficiency–equity frontiers trading off net social benefit against the concentration of gains in high-SVI populations.